

AMITH LAL

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Professional Summary

Results-driven AI and Machine Learning Engineer with specialized expertise in backend software engineering, full-stack application development, and deploying robust machine learning pipelines. Proficient in designing scalable RAG architectures, integrating local LLMs (Ollama), and building enterprise-grade automation using Python and Django. Proven track record of executing complex, large-scale data initiatives and seamless enterprise migrations to optimize cross-platform data flow and enhance overall business efficiency.

Education

APJ Abdul Kalam Technological University

Bachelor of Technology in Computer Science

Aug. 2019 – Jun. 2023

CGPA: 9.34

Experience

AI and Machine Learning Engineer

Jul 2024 – Present

Linux AS, Kochi

AI-Driven Document Processing & Extraction Platform | Python, Django, Ollama, LLMs, RAG

- Developed a comprehensive Django-based application utilizing **LLMs** via **Ollama** and **Retrieval-Augmented Generation (RAG)** to ingest, analyze, and extract structured data from diverse file formats (PDFs, Word, Excel, CSV, JSON).
- Interactive AI Capabilities:** Implemented **RAG** workflows allowing users to perform Q&A, generate summaries, and extract specific data points from uploaded files with high accuracy.
- SAP & API Integration:** Built robust Django REST APIs to handle file uploads and data extraction, integrating the solution directly with SAP systems for seamless enterprise data flow.
- Full-Stack Deployment:** Designed and hosted the application with a complete Django-based UI, enabling both API-driven and user-interface interactions for real-time document processing.

Albert Invent Data Migration & Utility Development | Python, Albert SDK, Pandas

- Executed complex data migration initiatives to transfer critical R&D data from diverse client legacy systems into the **Albert Invent** platform.
- Custom Utility Tooling:** Developed specialized utility tools using Python and the Albert SDK to automate data extraction, validation, and loading workflows.
- Effective Data Loading:** Engineered efficient scripts for bulk data processing, ensuring high data integrity and minimizing downtime during the migration capabilities.

Predictive Color Mixing Platform | Python, Machine Learning, Scikit-learn

- Formulation Optimization:** Engineered and deployed a predictive machine learning model using Scikit-learn to optimize industrial color mixing, significantly improving formulation accuracy and efficiency.

Cloud Engineering Intern

Feb 2024 – May 2024

Trainso Training Solutions, Bengaluru

- Worked on cloud infrastructure operations with a strong focus on AWS environments.
- Gained hands-on experience with IAM, EC2, S3, and basic security configurations.
- Assisted in deploying containerized applications and managing access control policies.

Certifications & Publications

- Machine Learning Specialization by Andrew Ng** – Coursera (2024)
- IBM Applied AI Professional Certificate** – Coursera (2023)
- Python for Data Science, AI & Development** – IBM via Coursera (2023)
- AWS Certified Solutions Architect Associate (SAA-C03)** – Udemy (2023)
- 30 Days of Google Cloud Program** – Google (2021)
- Research Publication:** *Personalizable Super-App Framework* – NCASCD (2023)

Technical Skills

- **Languages:** Python, SQL, Java
- **Machine Learning & AI:** Scikit-learn, Retrieval-Augmented Generation (RAG), MCP Servers, LangChain, Ollama, LLMs, FAISS, ChromaDB
- **Web, Backend & Automation:** Django, Django REST Framework, Firebase, n8n, Agentic Workflows, REST APIs
- **Cloud Platforms:** AWS (EC2, S3, IAM)
- **SAP Technologies:** ABAP, CDS Views, OData, FPM, Fiori, UI5
- **Databases & Storage:** Firestore, MySQL, SQLite
- **Version Control & Tools:** Git, GitHub, Postman, Figma, Jupyter, Adobe After Effects
- **Core CS Skills:** Data Structures and Algorithms, Object-Oriented Programming (OOP), API Integration